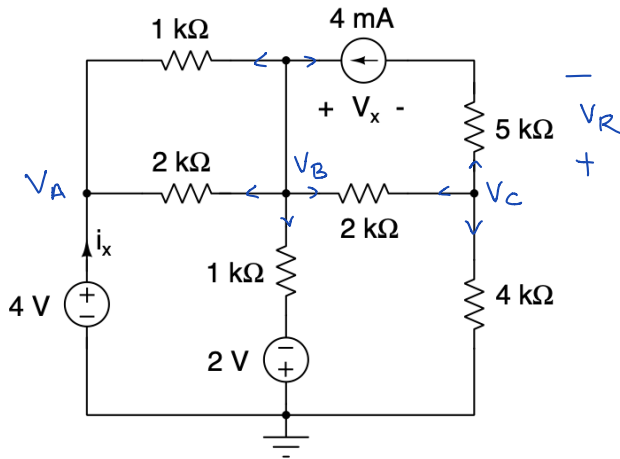


**EEE 211 - FALL 2021-2022
MIDTERM EXAM SOLUTIONS**

Question 1 [25 Points]:

- (a) Analyze the circuit given below using nodal analysis. Clearly label the node voltages (as V_A , V_B , V_C , ...). Find all the node voltages.
- (b) Using the result in part (a), find the current i_x and the voltage V_x as marked in the circuit.



a) $V_A = 4V$

KCL at V_B : $\frac{V_B - 4}{2k} + \frac{V_B + 2}{1k} + \frac{V_B - V_C}{2k} + \frac{V_B - 4}{1k} - 4mA = 0$

(2) (2) (2k)

$6V_B - V_C = 16 \Rightarrow V_C = 6V_B - 16 \dots \text{(Eq.1)}$

KCL at V_C : $\frac{V_C - V_B}{2k} + \frac{V_C}{4k} + 4mA = 0$

(2) (4k)

$3V_C - 2V_B = -16 \dots \text{(Eq.2)}$

Insert Eq.1 into Eq.2: $3(6V_B - 16) - 2V_B = -16$

$16V_B = 32 \Rightarrow \boxed{V_B = 2V}$

Insert into Eq.1: $V_C = 6 \cdot 2 - 16 = \boxed{-4V}$

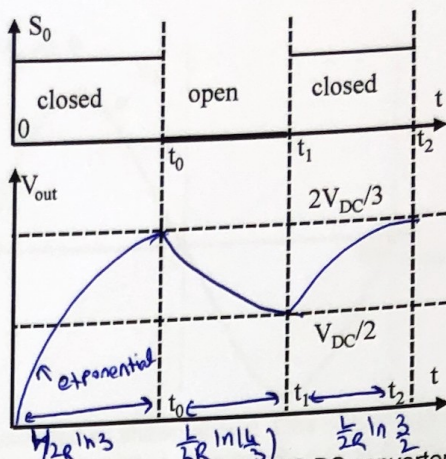
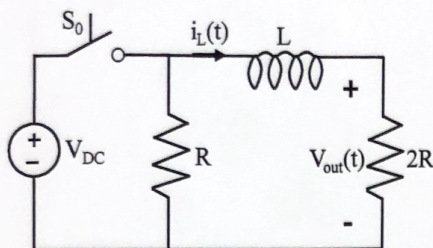
b) KCL at V_A : $-i_x + \frac{V_A - V_B}{2k} + \frac{V_A - V_B}{1k} = 0$

$i_x = \frac{4-2}{2k} + \frac{4-2}{1k} = \boxed{3mA}$

From KVL: $V_B - V_C = V_x - V_R$

$2 - (-4) = V_x - 4mA \cdot 5k \Rightarrow \boxed{V_x = 26V}$

Question 2 [25 points]:



The circuit shown above is called a Buck converter and is used as a step-down DC-DC converter. The switch S_0 is open (not conducting) for $t < 0$, and is then closed/opened/closed as marked in the graph above. We want $v_{out}(t_0) = v_{out}(t_2) = 2V_{DC}/3$ and $v_{out}(t_1) = V_{DC}/2$. Assume $i_L(0) = 0$.

Calculate the time intervals t_0 , $\Delta t_1 = (t_1 - t_0)$, and $\Delta t_2 = (t_2 - t_1)$ in terms of circuit parameters and plot $v_{out}(t)$ in the graph provided above.

for $0 - t_0$:

S_0 is closed and V_{out} will increase.

$$i_L(0) = 0 \quad V_{out}(0) = 0 \quad V_{out}(\infty) = V_{DC}$$

$$V_{out}(t) = V_f + (V_i - V_f)e^{-t/\tau}$$

$$V_{out}(t) = V_{DC} + (0 - V_{DC})e^{-t/\tau}$$

$$V_{out}(t) = V_{DC}(1 - e^{-t/\tau})$$

$$\tau = \frac{L}{2R}$$

$$V_{out}(t_0) = \frac{2V_{DC}}{3}$$

$$\frac{2V_{DC}}{3} = V_{DC}(1 - e^{-t_0/\tau})$$

$$e^{-t_0/\tau} = \frac{1}{3}$$

$$\ln(e^{-t_0/\tau}) = \ln(1) - \ln(3)$$

$$\frac{t_0}{\tau} = \ln 3$$

$$t_0 = \tau \ln 3$$

$$t_0 = \frac{L}{2R} \ln 3$$

$t_0 - t_1$, S_0 open $V_{out}(t)$ will decrease.

assume t_0 as our $t=0$ point

$$V_{out}(t_0) = \frac{2V_{DC}}{3} \quad V_{out}(\infty) = 0$$

$$V_{out}(t) = 0 + \left(\frac{2V_{DC}}{3} - 0\right)e^{-t/\tau}$$

$$\tau = \frac{L}{3R}$$

$$V_{out}(t) = \frac{V_{DC}}{2} = \frac{2V_{DC}}{3}e^{-t/\tau}$$

$$\frac{V_{DC}}{2} = \frac{2V_{DC}}{3}e^{-t/\tau}$$

$$e^{-t/\tau} = \frac{3}{4}$$

$$\frac{t_1}{\tau} = \ln\left(\frac{4}{3}\right) \Rightarrow t_1 = \frac{L}{3R} \ln\left(\frac{4}{3}\right)$$

$t_1 - t_2$: S_0 closed and $V_{out}(t)$ will increase.

assume t_1 as our $t=0$ point

$$V_{out}(t_1) = \frac{V_{DC}}{2} \quad V_{out}(\infty) = V_{DC}$$

$$V_{out}(t) = V_{DC} + \left(\frac{V_{DC}}{2} - V_{DC}\right)e^{-t/\tau}$$

$$V_{out}(t_2) = \frac{2V_{DC}}{3}$$

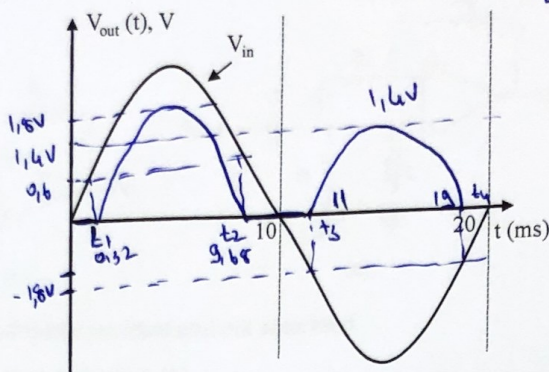
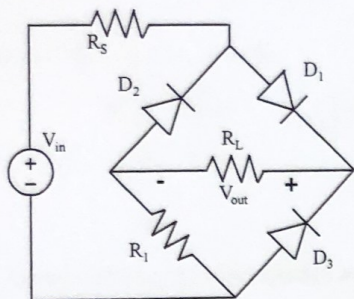
$$\frac{2V_{DC}}{3} = V_{DC} - \frac{V_{DC}}{2}e^{-t_2/\tau}$$

$$\frac{1}{2}e^{-t_2/\tau} = \frac{1}{3}$$

$$\frac{t_2}{\tau} = \ln\left(\frac{3}{2}\right)$$

$$t_2 = \frac{L}{2R} \ln\left(\frac{3}{2}\right)$$

Question 3 [25 points]:



for t_3 $1.8 = 6 \sin(2\pi 50 t)$
 $t = 9.68 \text{ ms} \sim 1 \text{ ms}$
 $t_3 = 11 \text{ ms}$
 $t_4 = 19 \text{ ms}$

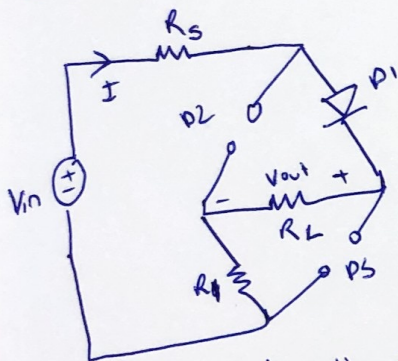
A full-wave rectifier circuit (without the capacitor) is modified as shown above: One of the diodes is replaced with a resistor (R_1) and a source resistor (R_s) is added. Assume $R_s = R_L = R_1 = R$, $V_{ON} = 0.6 \text{ V}$ for the diodes, and $v_{in}(t) = 6 \sin(2\pi 50 t) \text{ V}$.

Find $v_{out}(t)$ and plot it on the graph provided above, for one period. Mark important time points and voltages. Clearly label the peak voltage and diode turn on/off points.

Hint: Analyze the circuit separately for $v_{in}(t) > 0$ and $v_{in}(t) < 0$. You will need to write KCL for $v_{in}(t) < 0$ case.

$V_{in} > 0$

D_1 is ON, D_2 and D_3 are OFF



The current I can flow through R_s , D_1 , R_L and R_1

$I > 0$ when $V_{in} > V_{ON} = 0.6 \text{ V}$

$0 < V_{in} < 0.6 \text{ V}$ $V_{out} = 0$

$$V_{out} = \frac{R_L}{R_L + R_s + R_1} \cdot (V_{in} - 0.6) = \frac{1}{3} (V_{in} - 0.6)$$

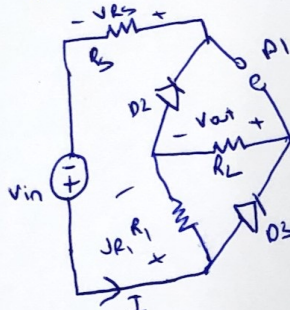
$$V_{out \text{ peak}} = \frac{1}{3} (6 - 0.6) = 1.8 \text{ V}$$

$$t_1 \rightarrow 0.6 = 6 \sin(2\pi 50 t_1)$$

$$t_1 = 0.32 \text{ ms}$$

$$t_2 = 10 - 0.32 = 9.68 \text{ ms}$$

$V_{in} < 0$ D_2 and D_3 are ON D_1 is OFF.



Note I flipped V_{in} .

when $V_{in} > 0.6 \text{ V}$

D_2 turns ON and the current starts flowing through R_1

but $V_{out} = 0$ since D_3 is OFF.

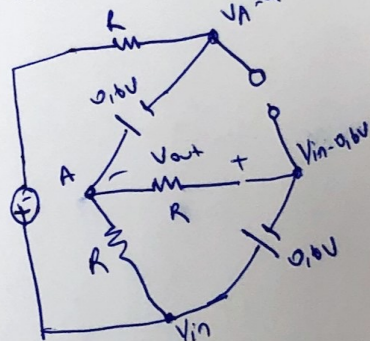
As V_{in} increases, D_3 turns ON

when $V_{R1} > V_{ON}$. @ that point

$V_{R_s} = V_{ON}$ so, $V_{out} = 0$ till

$V_{in} > 3V_{ON}$. V_{ON} V_{ON} V_{ON}

$V_{in} > 1.8 \text{ V}$ for D_3 to be ON.



for V_{out}

KCL @ A

$$\frac{V_A - V_{in}}{R} + \frac{V_A - (V_{in} - 0.6)}{R}$$

$$+ \frac{V_A - 0.6}{R} = 0$$

$$\frac{V_A - V_{in}}{R} + \frac{V_A - V_{in} + 0.6}{R} + \frac{V_A - 0.6}{R} = 0$$

$$3V_A = 2V_{in}$$

$$V_A = \frac{2}{3} V_{in}$$

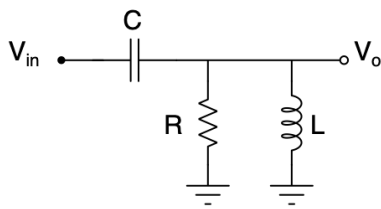
$$V_{out} = (V_{in} - 0.6) - \frac{2}{3} V_{in}$$

$$V_{out} = \frac{V_{in}}{3} - 0.6$$

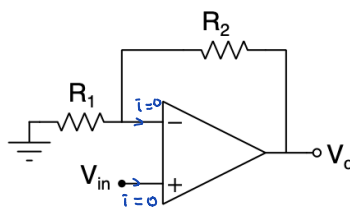
$$V_{out \text{ peak}} = \frac{6}{3} - 0.6 = 1.4 \text{ V}$$

$V_{in} > 1.8 \text{ V}$ for $V_{out} > 0$

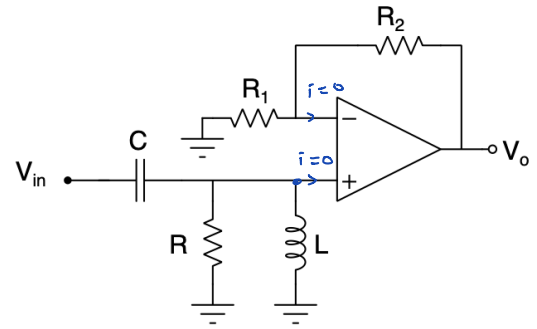
Question 4 [25 points]:



(a)



(b)



(c)

Assume that $v_{in}(t)$ is sinusoidal, and the OPAMPs are ideal and not saturated.

(a) Find the transfer function $H_1(\omega) = \frac{V_o}{V_{in}}$ for the circuit in (a).

(b) Find the transfer function $H_2(\omega) = \frac{V_o}{V_{in}}$ for the circuit in (b).

(c) Find the transfer function $H_3(\omega) = \frac{V_o}{V_{in}}$ for the circuit in (c).

(d) For the circuit in (c), assume that $R = 1 \text{ k}\Omega$, $R_1 = 2 \text{ k}\Omega$, $R_2 = 6 \text{ k}\Omega$, $C = 0.5 \text{ nF}$, $L = 1 \text{ mH}$. Find $v_o(t)$ if $v_{in}(t) = 2 \cos(10^6 t + \pi/4) \text{ V}$.

(e) For the same circuit in (d), find $v_o(t)$ if $v_{in}(t) = 2 \cos(10^7 t + \pi/4) \text{ V}$.

$$a) H_1(\omega) = \frac{V_o}{V_{in}} = \frac{R // j\omega L}{\frac{1}{j\omega C} + (R // j\omega L)} \quad (\text{voltage divider})$$

$$= \frac{\frac{R \cdot j\omega L}{R + j\omega L}}{\frac{1}{j\omega C} + \frac{R \cdot j\omega L}{R + j\omega L}} = \frac{j\omega L R \cdot j\omega C}{R + j\omega L - \omega^2 L C R} = \boxed{\frac{-\omega^2 L C}{(1 - \omega^2 L C) + j\omega \frac{L}{R}}}$$

$$b) V_+ = V_- = V_{in}$$

$$V_- = V_o \cdot \frac{R_1}{R_1 + R_2} \quad (\text{voltage divider})$$

$$\text{So, } V_{in} = V_o \cdot \frac{R_1}{R_1 + R_2} \Rightarrow H_2(\omega) = \frac{V_o}{V_{in}} = \boxed{1 + \frac{R_2}{R_1}}$$

$$c) V_+ \text{ is the same as } V_{out} \text{ in part (a): } \frac{V_+}{V_{in}} = H_1(\omega)$$

$$V_- \text{ is the same as } V_{in} \text{ in part (b): } \frac{V_o}{V_-} = H_2(\omega)$$

Using $V_+ = V_-$:

$$H(\omega) = \frac{V_o}{V_{in}} = \frac{V_+}{V_{in}} \cdot \frac{V_o}{V_-} = \boxed{\frac{R_1 + R_2}{R_1} \cdot \frac{(-\omega^2 L C)}{(1 - \omega^2 L C) + j\omega \frac{L}{R}}}$$

Note: Here, we see the buffering property of the op-amp. It isolates the input side from the output side.

$$d) \omega = 10^6 \text{ rps}, \quad V_{in} = 2 \cdot e^{j\pi/4} \text{ V}$$

$$H(\omega) = \frac{2k+6k}{2k} \cdot \frac{(-10^{12} \times 10^{-3} \times 0.5 \times 10^{-9})}{(1 - 10^{12} \times 10^{-3} \times 0.5 \times 10^{-9}) + j 10^6 \cdot \frac{10^{-3}}{10^3}}$$

$$= 4 \cdot \frac{(-0.5)}{(1 - 0.5) + j} \stackrel{(2)}{=} \frac{-4}{1 + 2j} = \frac{4 \cdot e^{j\pi}}{\sqrt{5} e^{j\alpha \tan(z)}}$$

$$V_{out} = H(\omega) \cdot V_{in} = \frac{4 \cdot e^{j\pi}}{\sqrt{5} e^{j\alpha \tan(z)}} \cdot 2 e^{j\pi/4}$$

$$= \frac{8}{\sqrt{5}} \cdot e^{j(5\pi/4 - \alpha \tan(z))}$$

$$V_{out}(t) = \frac{8}{\sqrt{5}} \cos(10^6 t + 5\pi/4 - \alpha \tan(z)) \text{ V}$$

$$V_{out}(t) = 3.58 \cos(10^6 t + 2.82) \text{ V} = 3.58 \cos(10^6 t + 161.6^\circ) \text{ V}$$

$$e) \omega = 10^7 \text{ rps}, \quad V_{in} = 2 \cdot e^{j\pi/4} \text{ V}$$

$$H(\omega) = \frac{2k+6k}{2k} \cdot \frac{(-10^{14} \times 10^{-3} \times 0.5 \times 10^{-9})}{(1 - 10^{14} \times 10^{-3} \times 0.5 \times 10^{-9}) + j 10^7 \cdot \frac{10^{-3}}{10^3}}$$

$$= 4 \cdot \frac{-50}{(1 - 50) + j10} = \frac{200}{49 - j10} = \frac{200}{\sqrt{49^2 + 10^2}} e^{-j\alpha \tan(\frac{10}{49})}$$

$$V_{out} = H(\omega) \cdot V_{in} = \frac{200 \cdot 2 \cdot e^{j\pi/4}}{\sqrt{49^2 + 10^2}} e^{-j\alpha \tan(\frac{10}{49})} = \frac{400}{\sqrt{49^2 + 10^2}} e^{j(\pi/4 + \alpha \tan(10/49))}$$

$$V_{out}(t) = \frac{400}{\sqrt{49^2 + 10^2}} \cos(10^7 t + \pi/4 + \alpha \tan(10/49)) \text{ V}$$

$$V_{out}(t) = 8 \cos(10^7 t + 0.99) \text{ V} = 8 \cos(10^7 t + 56.5^\circ) \text{ V}$$